

Keep the planner in the curriculum.

Not on the robot. BEHAVIOR-class homes are obstructed and full of contact. MoPA-PD's idea is a teacher/student split — not a Sawyer checkpoint, and not a baseline we are inventing.

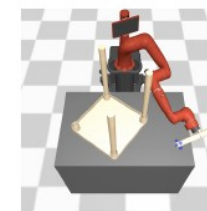
No Discord. No Drive. No unofficial BEHAVIOR score. The 2021 scenes are a box, a can, and a table leg. The analogy has to survive that modesty.



(a) Sawyer Push



(b) Sawyer Lift



(c) Sawyer Assembly

Five facts that do not change with the seat

Train privilege

01

Classical MP (RRT-Connect) lives inside MoPA-RL (Yamada et al., CoRL 2020). Geometry and the planner are train-time infrastructure, not the shipped controller.

Distill, then smooth

02

Visual BC on direct-action rollouts removes the planner and smooths RRT jitter. MoPA-PD then runs asymmetric SAC: actor from BC, critic from MoPA-RL Q , $Re:R\pi = 1:3$, small α .

Deploy is planner-free pixels

03

Inference is a 32×32 image plus joints. No RRT-Connect, no object state, no motion planner. The critic's state privilege ends when training ends.

Numbers you must be able to say

04

Push 100 / 32.0 / 110.8 (ASR / AEL / ADR). Assembly 0 without BC-smooth vs 100 with. $DR \geq 96.7$ sim-to-sim (Lift Scenario 1 is worst). Expert Push AEL 111 vs Ours 32.

What this paper is not

05

Not an ancestor of ACT, Diffusion Policy, π_0 , OpenVLA, or OXE — contrast poles only. Not a JEPA / LeWM / Cosmos child. Sim-only Sawyer. Do not launder DR into sim-to-real.

Privilege is a curriculum, not a sensor

When the mesh is known, let a planner explore the clutter. Distill until a visual policy no longer needs the mesh. Then delete the planner from the runtime.

Known in sim

Collision geometry, object poses, a motion planner or a TAMP teacher. Stage 0. It does not ship.

Unknown at home

Cameras and proprioception. That is the actor. State critics and RRT stay in the training room.

The 2021 proof

Three Sawyer tasks, success near 100, paths shorter than the teacher, Assembly impossible without smoothing.

The 2026 limit

Analogy only. No BEHAVIOR success rate is in this paper. Do not speak as if a leaderboard recorded one.

What the pattern suggests. What it does not score.

Collision-heavy rooms

Train-time RRT, or a TAMP teacher, for reach and place. Not online on the robot.

Vision-only deploy

Asymmetric train, visual actor at test. A vision critic is future work the authors already named.

Multi-stage contact

Lift is already reach, grasp, lift. Household tasks need stage-wise teachers, then a smooth, then a modern head.

Appearance shift

Keep domain randomization. Table 2 is the evidence: textures and distractors, not a new kitchen.

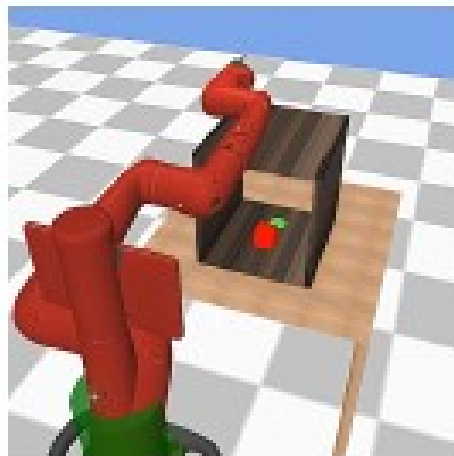
Layout shift

Not tested. Rearranged furniture as kinematic OOD is outside Table 2. Say the gap.

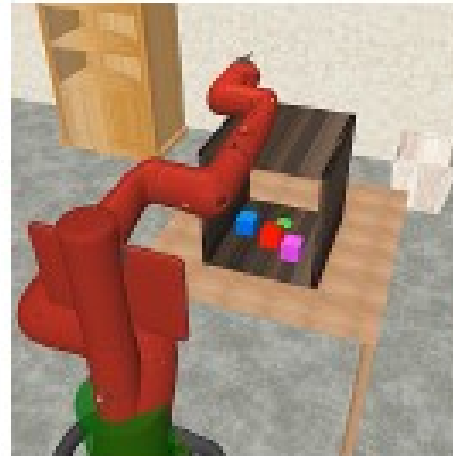
Distractors in a room the simulator already knows



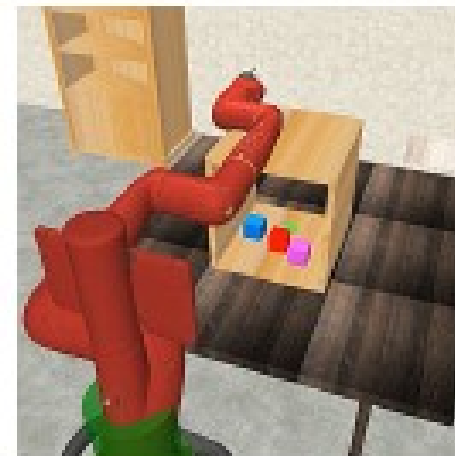
(a) Domain randomization



(b) Original environment



(c) Scenario 1



(d) Scenario 2

Inside the claim

Unseen colors, furniture, table size, box texture. ASR at least 96.7. The authors call it sample-efficient.

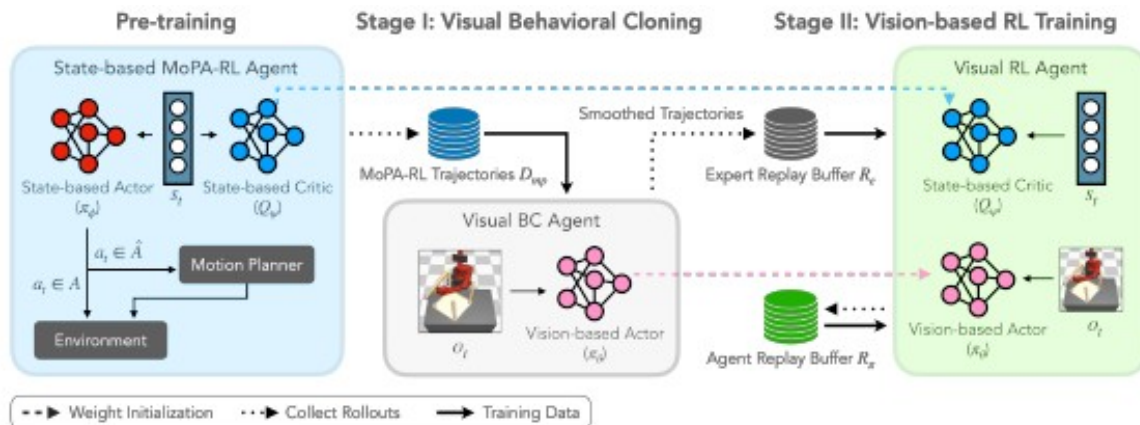
Outside the claim

New kinematics, novel geometry, real cameras, force, language goals. Not measured. Not “almost.”

Curriculum use

Appearance DR stays, as a cheap invariance. Add layout randomization on purpose. Do not cite Table 2 as having done it.

TENTATIVE. A data role. Not an architecture merger.



TENTATIVE

Planner-teacher demos can be one column beside teleop.

The head can be BC, ACT, diffusion, or a VLA fine-tune.

The 32×32 CNN does not come along.

Must keep

Smooth before you mix. Raw RRT in a modern buffer repeats the Assembly failure, whatever the head.

Must drop

32×32 SAC as the 2026 controller. The substrate changes. The teacher/student split does not have to.

Must not say

That this replaces a VLA, or that those models descend from this figure.

Modern heads stay modern. The teacher is the new column.

Do not replace the head

Chunking, diffusion, flow, and language solve interfaces MoPA-PD never had. Use them. Feed them better clutter data.

Do not replace the teacher

Human teleop under-samples collision-free transit through tight geometry. A privileged planner is one way to cover that slice. Not the only column.

Do not ship RRT online

Unless a latency budget truly allows it. The point of this paper is amortization. A robot that calls RRT on every large motion missed the lesson.

Do not erase the critic

A state value in sim is allowed. A state value at a real eval is a different system. The authors already asked for a vision critic first.

MoPA-PD did not invent the family. It is a clean 2021 member.

Asymmetric critics

Pinto et al. 2018, used here directly. State critic, visual actor. Still common in sim. Still not a free real-world critic.

Teacher to student

A planner or GT state rolls out; the student clones or is fine-tuned. MoPA-PD adds smoothing and a low- α exploit bias on top.

DR, then a small real step

Tobin-style randomization, then zero-shot or a light fine-tune. This paper stops at zero-shot sim. The real step is unnamed future work.

Primitive plus residual

RelMoGen's bucket, and later MP menus in the same family. Contrast: they keep the primitive available. MoPA-PD deletes it at deploy.

Six directions. Zero fake baselines.

- 01** Swap the 32×32 CNN for a modern encoder. Keep the MP teacher. Measure whether smoothing still matters.
- 02** Train the vision critic the conclusion asks for. That handoff is the real-robot blocker, not a slogan.
- 03** Store teacher demos as shards tagged with the teacher. One column in a larger set, not a new dataset religion.
- 04** Try BC-smooth in front of action chunking. TENTATIVE. Assembly says jitter is poison; chunking does not repeal it.
- 05** Equal hours of privileged-MP demos versus teleop on obstruction-heavy tasks. A future experiment. Not a known winner.
- 06** Treat planner jitter as a data filter before IL. Fig. 8 is the picture. A numeric cutoff is something you would still have to define.

Applications as hypotheses

Inserts among obstacles

Assembly is already an insert next to fixed legs. Cabinet curricula in sim are the straight extension. Still a hypothesis.

Reach behind clutter

Fridge, shelf, box. Push and Lift are the 2021 cartoons of that problem. A planner covers the approach. It does not replace touch at the grasp.

Hard-negative layouts

Use the teacher to build obstacle arrangements that visual policies fail. An eval set you might make. Not a claimed BEHAVIOR split.

Latency-critical deploy

Bake the planner's cost into a reactive policy so the robot does not wait. That is §1's motivation, and still the right product sentence.

Imagination ends where the PDF ends

No Discord. No Drive.

Nothing in this seat is a message, a folder, or an upload. The deliverable is the argument.

No invented baseline

No official BEHAVIOR '26 MoPA-PD entry is in the sources. Do not promote the paper onto a leaderboard it did not enter.

No invented robot SR

Real-robot transfer is future work. DR is sim-to-sim. A percent you did not read is a fiction.

No false family

Not the parent of ACT, Diffusion Policy, π_0 , OpenVLA, OXE, LeWM, JEPa, or Cosmos. A teacher-role rhyme is the strongest legal claim.

Bring Table 1's metrics. Leave Table 1's scores.

A household distill is not done at success rate. The 2021 flex was a shorter, higher-return path than the privileged expert — and a total miss when the buffer was jittery.

Stage success

Reach, grasp, transport, insert. Lift's reward already has three stages. Log them.

Path and collisions

AEL's moral. A policy that succeeds by thrashing repeats the expert's failure mode.

Privilege audit

Planner calls at deploy = 0. State in the critic = train only. Same demand a reviewer would make.

Smoothing audit

Jerk of teacher demos vs smoothed demos vs the deployed policy. Assembly shows this is not cosmetic.

The teacher can leave the runtime

For obstructed homes, keep classical planning as a sim teacher. Distill — smooth first — into whatever vision student you actually trust. The 2021 proof is that the teacher can disappear at deploy without disappearing from the curriculum.

Push 100 / 32.0 / 110.8. Assembly 0 without the smooth. $DR \geq 96.7$, sim only. Not a VLA ancestor. Not a world model. No score we did not read.